RAPID DIC50 CALCULATION

Well Plate analysis from 96 well plate images

This application enables researchers to rapidly calculate the half-maximal inhibitory concentration (IC_{50}) of compounds tested in a standard 96-well cell culture plate. Two workflows are provided: AI-powered image analysis (dIC_{50}) and manual data entry.

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1. INTRODUCTION TO THE APPLICATION

1.1 What is this Application?

The 96-Well Plate dIC_{50} Analyser is a browser-based web application specifically designed for biological and pharmaceutical researchers who need to quickly and accurately determine the inhibitory concentration at 50% cell viability (IC_{50}) for drug compounds tested in standard 96-well culture plates.

Traditional IC_{50} analysis often requires expensive plate reader equipment and complex software pipelines. This application replaces that process with an intuitive, AI-powered pipeline that works directly from a photograph of your 96-well plate, taken with any standard camera or smartphone. It also supports fully manual data entry for researchers who already have viability measurements from other instruments.

Think of it this way: instead of feeding your plate into an expensive reader machine, you simply take a clear photograph of it, upload it to this app, point at three wells to calibrate the grid, assign your control and concentration wells, and hit a button. The app does the rest reading each well's color intensity, normalizing it against your controls, and instantly producing a dose-response curve with the calculated IC50.

1.2 What is IC_{50} and dIC_{50}?

IC_{50} stands for Half Maximal Inhibitory Concentration. It is one of the most important measurements in pharmacology and toxicology. It tells you: "How much of this drug compound do I need before 50% of the cells stop living (or responding)?"

A lower IC50 means the drug is more potent you need less of it to achieve the same effect. A higher IC50 means you need more of the compound. Scientists use this number to compare different drugs, formulations, and target cells.

dIC_{50} (Digital IC_{50}) is the term used in this application for IC_{50} values that are derived computationally from digital image analysis, as opposed to traditional spectrophotometric readings. The "d" prefix distinguishes results obtained by this image-based pipeline from those obtained by physical plate readers.

Concept	Plain English Explanation
96-Well Plate	A small plastic tray with 96 tiny cups (wells) arranged in 8 rows × 12 columns. Each well can hold a small volume of liquid with cells and drug.
Cell Viability (%)	The percentage of cells that are still alive. 100% = all alive, 0% = all dead.
IC_{50}	The drug dose at which 50% of cells have died or stopped functioning.
dIC_{50}	The IC50 calculated digitally from a photo of your well plate, using this software.
Dose-Response Curve	A graph showing how increasing the drug dose reduces cell viability. The IC50 is the concentration at the 50% viability mark.

1.3 Who Should Use This Tool?

- Pharmacology Researchers**:** Testing new drug compounds for potency against cell lines.
- Toxicologists: Measuring cytotoxicity of chemicals or environmental samples.
- Biotechnology Students: Learning IC_{50} analysis as part of major or thesis projects.
- Laboratory Scientists: Needing rapid, equipment-free IC_{50} estimation from plate photos.
- Data Analysts: Processing pre-measured viability data via the manual entry workflow.

2. GETTING STARTED

2.1 System Requirements

The application runs entirely in your web browser. No installation is required. However, your system should meet the following minimum requirements:

Requirement	Minimum	Recommended
Web Browser	Chrome 90+, Firefox 88+, Edge 90+, Safari 14+	Latest version of Google Chrome
Internet Connection	Required for initial load	Stable broadband
Screen Resolution	1024 × 768	1920 × 1080 or larger
RAM	4 GB	8 GB or more
Camera/Image	Any smartphone camera	DSLR or lab camera with even lighting
Image Format	PNG, JPG, GIF	PNG or high-quality JPG, up to 10 MB

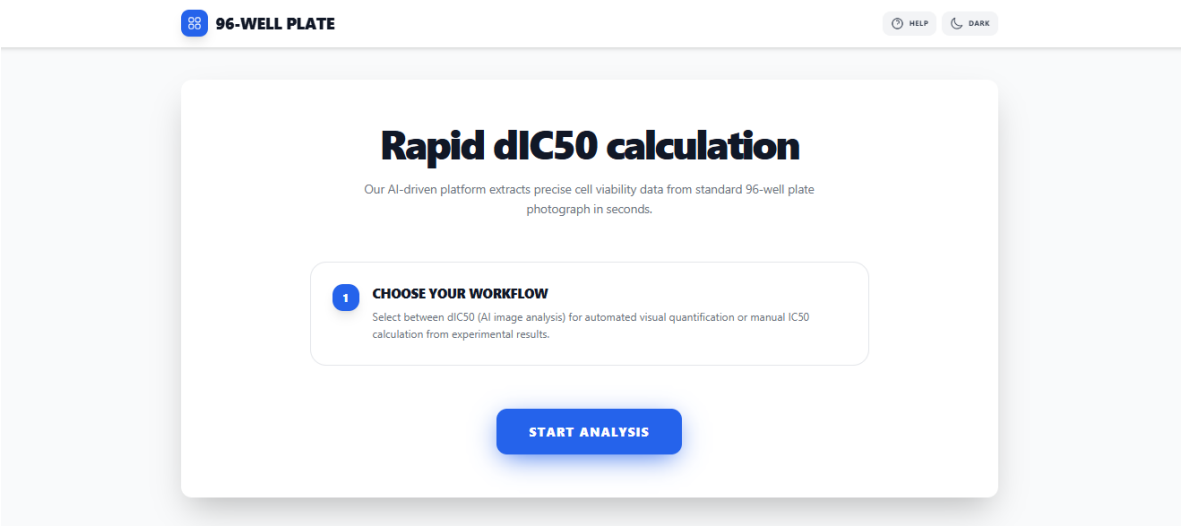
2.2 Opening the Application

Open your web browser and navigate to the application URL provided by your instructor or institution. The application will load automatically. No login or account creation is required. All processing happens locally in your browser your images and data are never uploaded to any external server.

2.3 Interface Overview

The application features a clean, modern interface with the following persistent elements visible at all times:

- Header Bar: Always visible at the top. Contains the app logo, title "96-WELL PLATE", a Back button (when navigating), a Help button, and a Dark/Light mode toggle.
- Main Content Area: The large white/dark card in the center. This area changes depending on which step you are on.
- Footer: A minimal footer at the very bottom of the screen.



3. WORKFLOW A: dIC_{50} AI IMAGE ANALYSIS

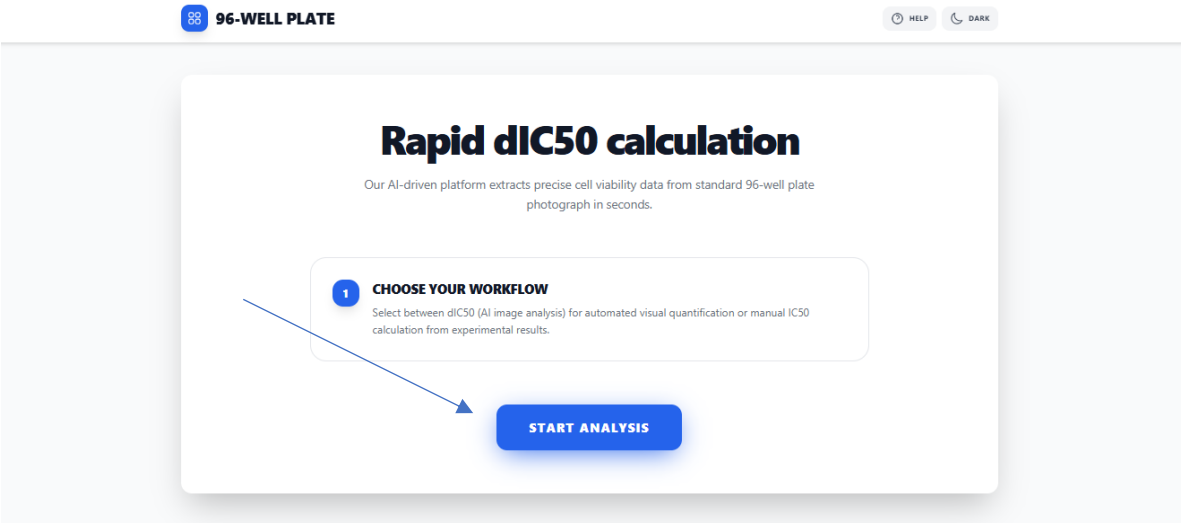
This is the primary workflow of the application. It uses your plate photograph and AI-driven color analysis to automatically measure cell viability in each well and compute the IC_{50} value. Follow each step carefully for best results.

3.1 Step 1 - The Welcome Page

When you first open the application, you will see the Welcome Page. This page explains the core workflow of the tool in a brief overview.

It contains a large heading reading "Rapid dIC_{50} calculation", a short description of the platform, and a step overview card showing the workflow steps.

At the bottom there is a large button labelled "Initialize Workspace". Click this button to begin.



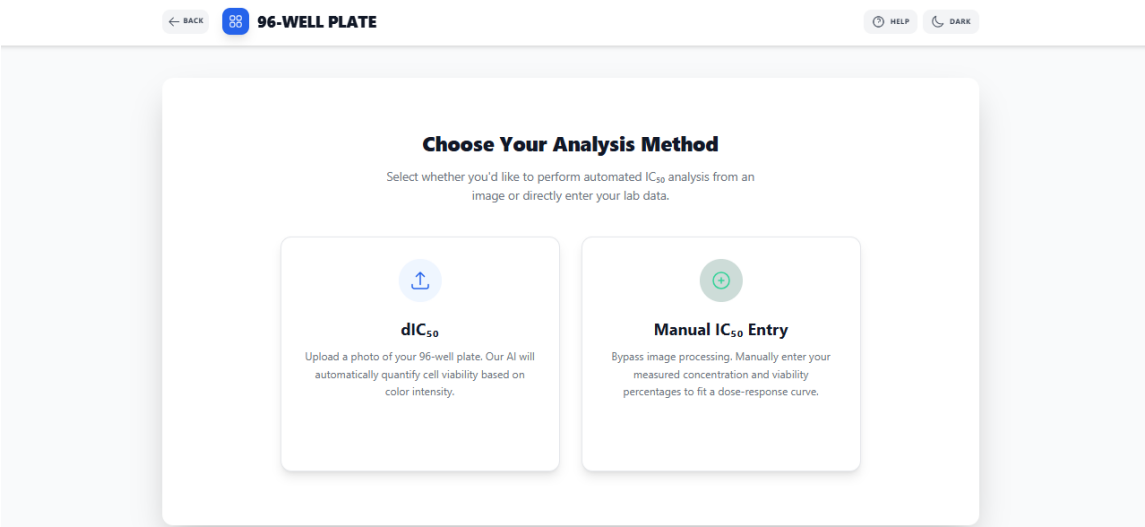
3.2 Step 2 - Choosing Your Analysis Method

After clicking "Initialize Workspace", you will arrive at the Method Selection Page. Here you choose between two analysis pathways. This section describes the dIC_{50} pathway.

You will see two large cards side by side:

- dIC_{50} (left card): For AI-powered image analysis. Click this card to upload a photograph of your well plate and let the app measure viability automatically.
- Manual IC_{50} Entry (right card): For manual data entry. Skip to Section 4 of this manual for instructions on this path.

Click the dIC_{50} card to proceed with the image analysis workflow. When you hover over the card, it highlights and shows an "Upload Image" button.

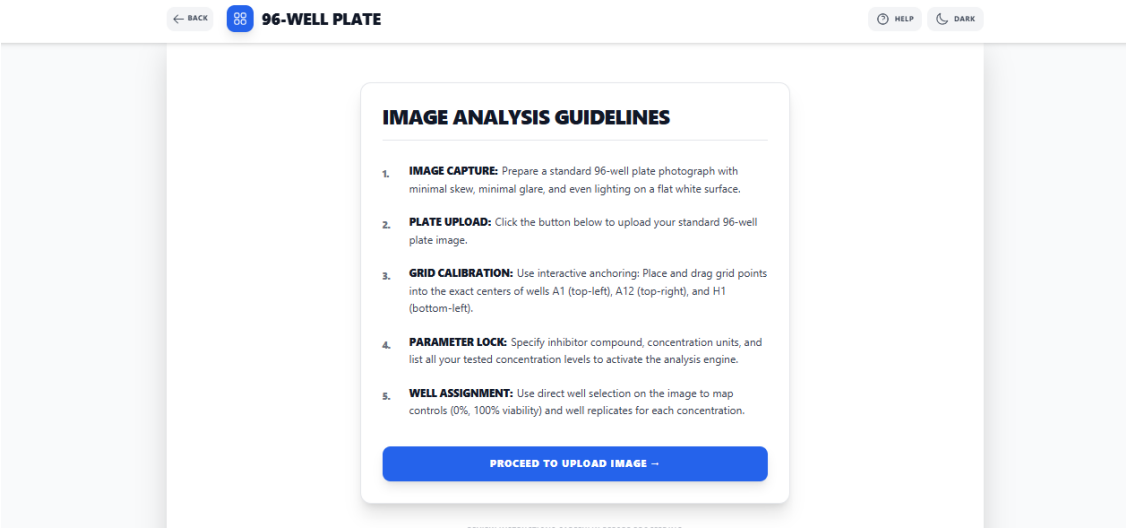


3.3 Step 3 - Reading the Guidelines

After selecting dIC_{50}, the app shows you the Image Analysis Guidelines page. This is an important page that explains how to correctly prepare your sample and image before proceeding. Read each guideline carefully.

- IMAGE CAPTURE: Photograph your 96-well plate on a flat white surface with even, diffuse lighting. Avoid glare, shadows, or strong directional light. The plate should be as straight (not tilted) as possible in the frame.
- PLATE UPLOAD: Click the "Proceed to Upload Image" button to go to the upload screen where you will select your image file.
- GRID CALIBRATION: You will interactively click on three corner wells (A1 top-left, A12 top-right, H1 bottom-left) to define the grid over your plate.
- PARAMETER LOCK**:** You will enter your inhibitor compound name, concentration units, and all concentration levels tested.
- WELL ASSIGNMENT: You will click on individual wells in the image to assign them as controls (0% or 100% viability) or concentration wells.

At the bottom of the page is the "Proceed to Upload Image →" button. Click it when you are ready.

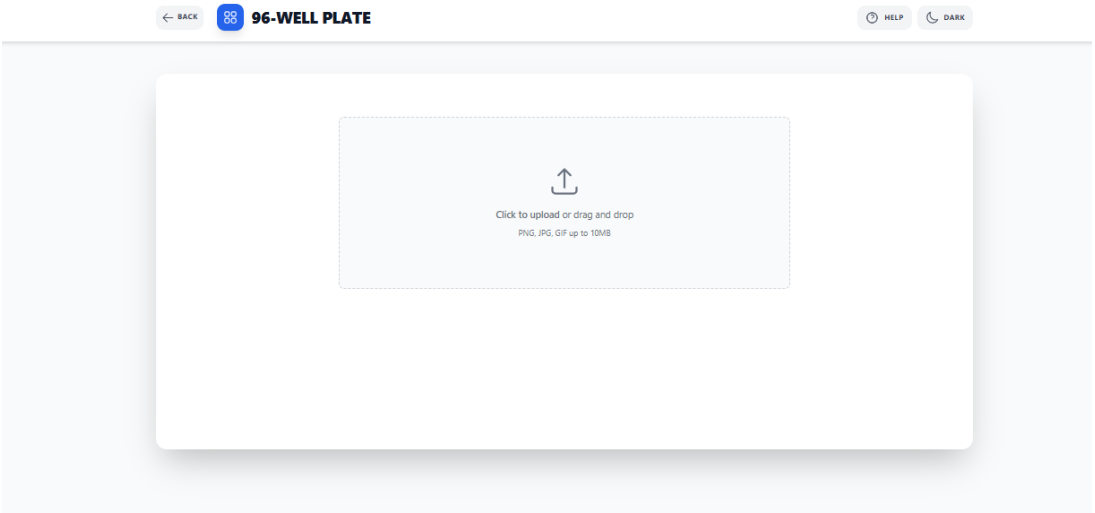


3.4 Step 4 - Uploading Your Plate Image

The **Image Upload Screen** presents a large dashed upload area. You can add your image in two ways:

Method A Click to Browse: Click anywhere on the dashed box. A standard file browser window will open. Navigate to your image file (PNG, JPG, or GIF, up to 10 MB) and select it.

Method B Drag and Drop**:** Open your file manager, find your image, and drag it directly onto the dashed box. The border will change colour to indicate the drop zone is active. Release the file to upload it.

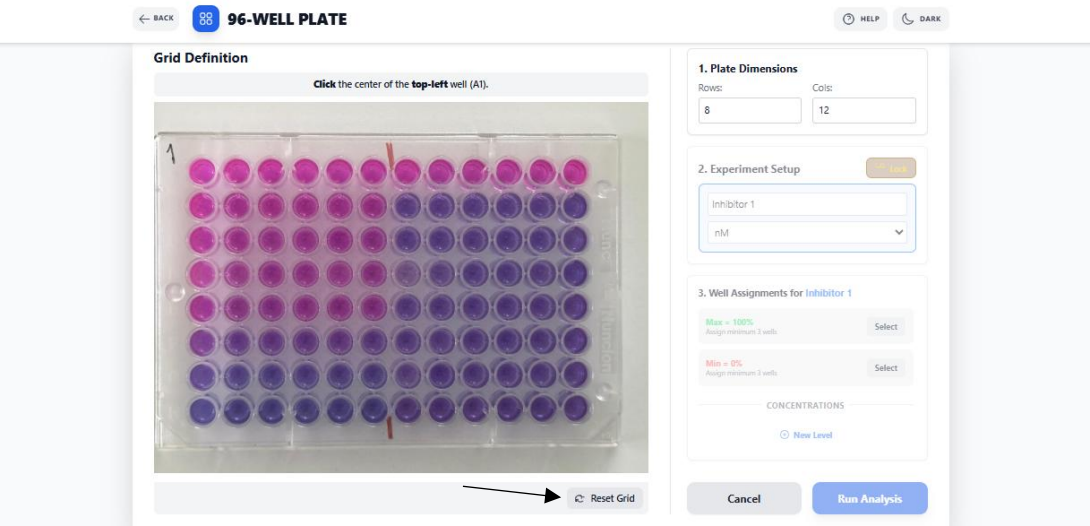


Once you select a valid image file, the application will automatically advance to the Grid Calibration screen. No extra button press is needed.

3.5 Step 5 - Preparing Your Image (Optional: Crop & Rotate)

After your image loads on the Grid Definition screen, you will see it displayed on the left side of the page. If necessary, you can adjust the image before calibrating the grid using the toolbar below the image:

- **Reset Grid button:** Clears all calibration points and any cropped or rotated image modifications, returning you to the original uploaded image.



3.6 Step 6 - Calibrating the Well Grid

Grid calibration is the most important step. You are telling the application *exactly* where your wells are located within the image. This is done by clicking on the centers of three specific wells, one at a time.

A blue instruction box above the image guides you through each sub-step:

Sub-step 1: Click Well A1 (Top-Left)

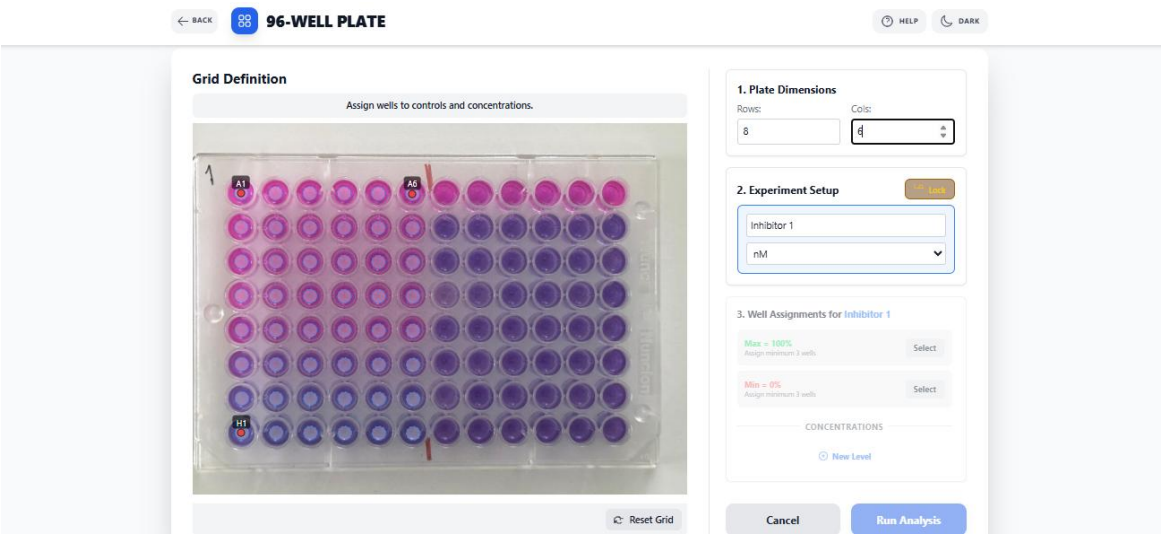
The app prompts: "Click the center of the top-left well (A1)." This is the very first well in the top-left corner of the plate (Row A, Column 1). Click as precisely as possible on the center of this circular well. A small red dot with the label "A1" will appear to confirm your click.

Sub-step 2: Click Well A12 (Top-Right)

The app advances automatically and prompts: "Click the center of the top-right well (e.g., A12)." This is the last well in the top row. Click its center. A second red dot appears labelled "A12".

Sub-step 3: Click Well H1 (Bottom-Left)

Finally, the app prompts: "Click the center of the bottom-left well (e.g., H1)." This is the first well in the bottom row. Click it. A third red dot appears labelled "H1". The app will now automatically snap the full grid overlay to your plate and advance to the Well Assignment step.



3.7 Step 7 - Setting Up the Experiment (Right Panel)

The right side of the Grid Definition screen contains three setup panels that must be completed before running the analysis. They become enabled progressively.

Panel 1: Plate Dimensions

This panel shows input fields for Rows and Cols (columns). By default these are set to 8 rows and 12 columns, which is correct for a standard 96-well plate. If you are using a different plate format (e.g., 24-well or 48-well), change these values before calibrating the grid.

Panel 2: Experiment Setup

This panel becomes active after you place the third grid calibration point. It is used to enter information about your inhibitor compound. It contains:

- Inhibitor Name field**:** Type the name of your (eg: sorafenib)
- Concentration Units dropdown: Select the units for your tested concentrations: nM (nanomolar), µM (micromolar), or mM (millimolar).

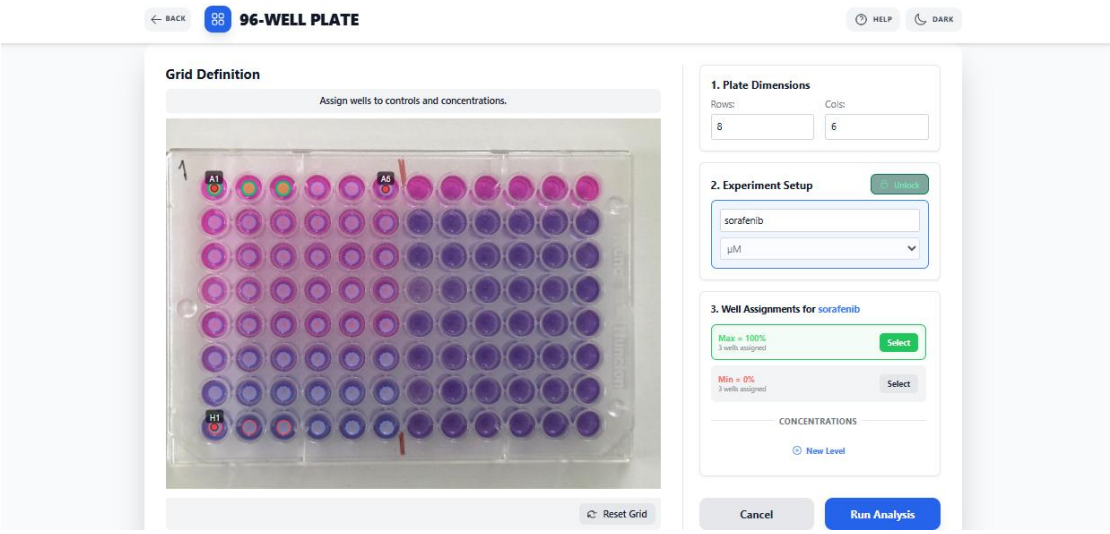
Once you have filled in the inhibitor name and units, click the "Lock" button (shown with a padlock icon). This locks the experiment parameters and activates the Well Assignment panel below. You can unlock at any time to make changes.

Panel 3: Well Assignments

This is where you tell the application which wells correspond to your controls and which correspond to each tested drug concentration. This panel has three sections:

- Max = 100% (Untreated / Positive Control): These are your wells containing cells with no drug treatment — the highest viability reference. Click "Select" next to this row to activate it (it turns green), then click on each well in the plate image that you want to assign as an untreated control. Assign at least 3 wells. The count updates as you click.

- Min = 0% (Blank / Negative Control): These wells contain no living cells just medium representing the lowest viability (background signal). Click "Select" for this row, then click your blank/media-only wells on the image. Assign at least 3 wells.
- Concentrations section: Below the controls, add each concentration level you tested. Click "New Level" to add a concentration row. Type in the concentration value (number only, in the units you selected). Then click the orange plus button or click the row to activate it, and click the corresponding wells on the plate image. Repeat for each concentration. Each well you click turns orange and shows the concentration value inside it.



3.8 Step - Assigning Concentrations to Wells

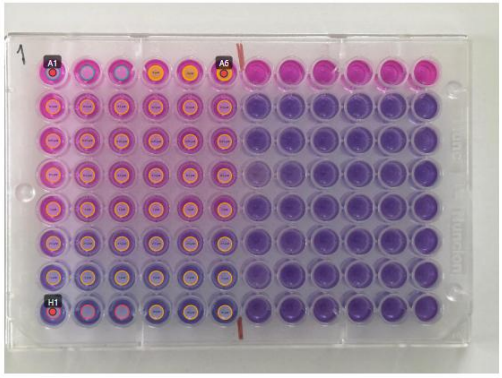
Before running the analysis, concentration values must be assigned to the selected wells. This step defines the dose-response relationship required for IC_{50} calculation.The application provides an interface to input concentration levels and map them to corresponding wells on the plate.

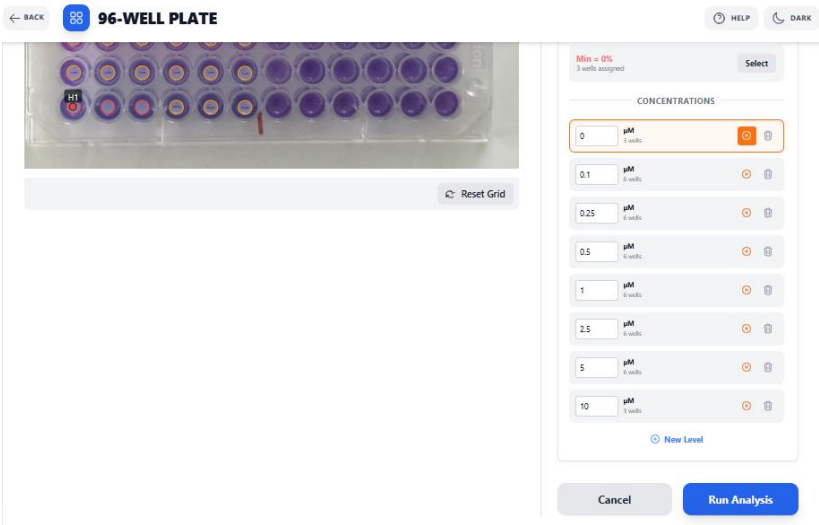
Procedure-

1. **Access the Concentration Panel:**
   Navigate to the “Concentrations” section available on the right-side panel under experiment setup.
2. **Enter Concentration Values:**
   Input the desired concentration value (e.g., 0, 10, 50) and select the appropriate unit (e.g., µM, mM).
3. **Add Multiple Concentration Levels:**
   Use the *“New Level”* option to define multiple concentration groups corresponding to different doses.
4. **Assign Wells to Concentrations:**
   For each concentration level:
   - Click the “Select” button
   - Choose the wells directly from the plate image
   - Assigned wells are visually highlighted for confirmation
5. **Verify Assignments:**

Ensure that:

- Each concentration level has assigned wells
- Control groups are correctly defined
- No well is assigned to multiple conflicting groups





3.9 Step 8 - Running the Analysis

Once you have calibrated the grid, locked the experiment setup, and assigned at least one concentration with its wells (plus both control groups), the blue "Run Analysis" button at the bottom right becomes active.

Click "Run Analysis" to begin. The app will:

1. Lock the image at a standardised 1024-pixel wide resolution for consistency.
2. Read the average pixel color inside each assigned well using a circular region ofinterest (ROI).
3. Convert colors to the perceptually-uniform CIE L*a*b* color space for accuratecomparison.
4. Calculate the color gradient from your 0% control (min) to 100% control (max).
5. Project each concentration well's color onto this gradient to compute a viabilitypercentage.

6. Apply a camera calibration correction (gamma correction factor) to matchspectrophotometer physics.
7. Calculate average viability, standard deviation, and standard error for eachconcentration group.
8. Perform linear interpolation between data points to find the IC50.
9. Display the Results page with all findings.

A spinning loading indicator appears while the computation runs. This typically takes less than a second for standard 96-well plates.

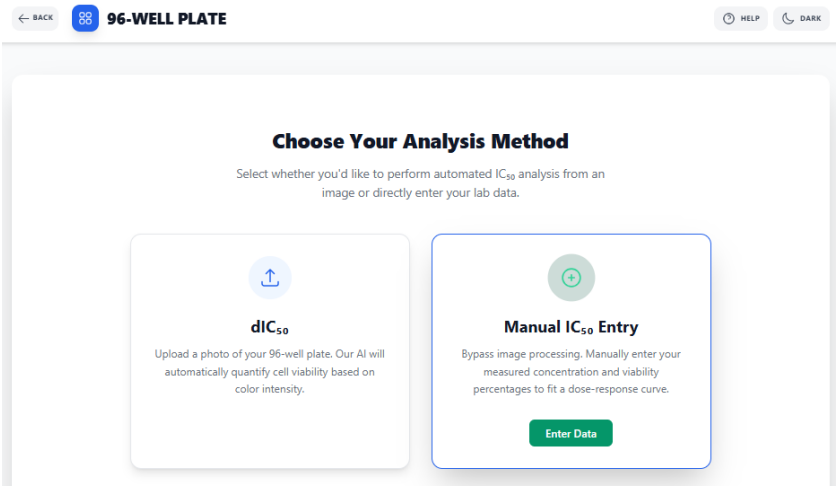
4. WORKFLOW B: MANUAL IC_{50} ENTRY

This workflow is designed for researchers who already have cell viability measurements from a plate reader or other instrument and want to use the application's curve-fitting and IC_{50} calculation engine without uploading an image.

4.1 Step 1 - Choose Manual Entry

On the Method Selection page, click the **green** "Manual IC_{50} Entry" card on the right side. You will be shown the Manual Data Entry Guidelines page, which explains what to prepare before proceeding.

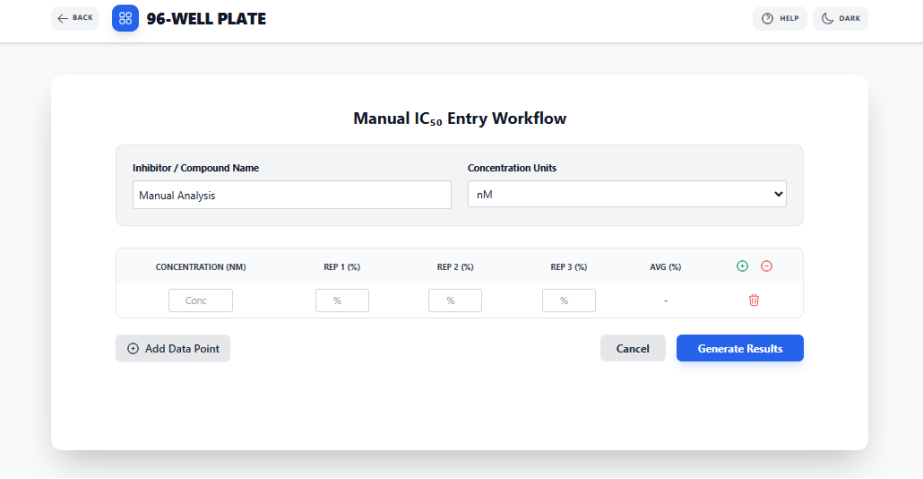
Click "Proceed to Data Entry →" at the bottom of the guidelines page.



4.2 Step 2 - Setting Up Parameters

The Manual Entry Workflow screen opens. At the top there is a two-column settings area:

- Inhibitor / Compound Name**:** Type the name of your compound. This name will appear on graphs and in exported files.
- Concentration Units**:** Select from the dropdown: nM, µM, or mM. This label will appear on the X-axis of the dose-response graph.



4.3 Step 3 - Entering Concentration and Viability Data

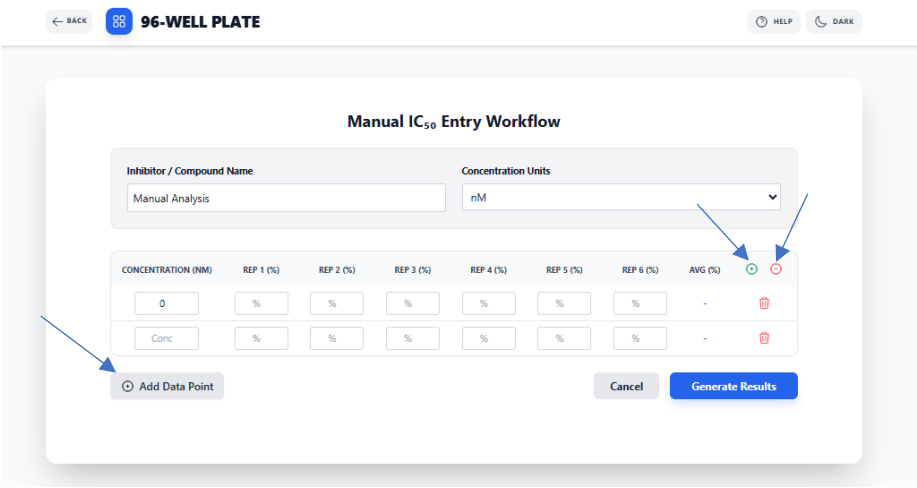
Below the settings is the **data entry table**. Each row represents one concentration level. The columns are:

Column	What to Enter	Example
Concentration	The numeric value of the drug concentration for that row.	10 (meaning 10 nM)
Rep 1, Rep 2, Rep 3...	Cell viability percentage for each replicate well at that concentration.	85.2, 87.1, 84.9
Avg (%)	Auto-calculated average of all replicate values in that row.	85.7 (auto)

4.4 Step 4 - Managing Rows and Replicates

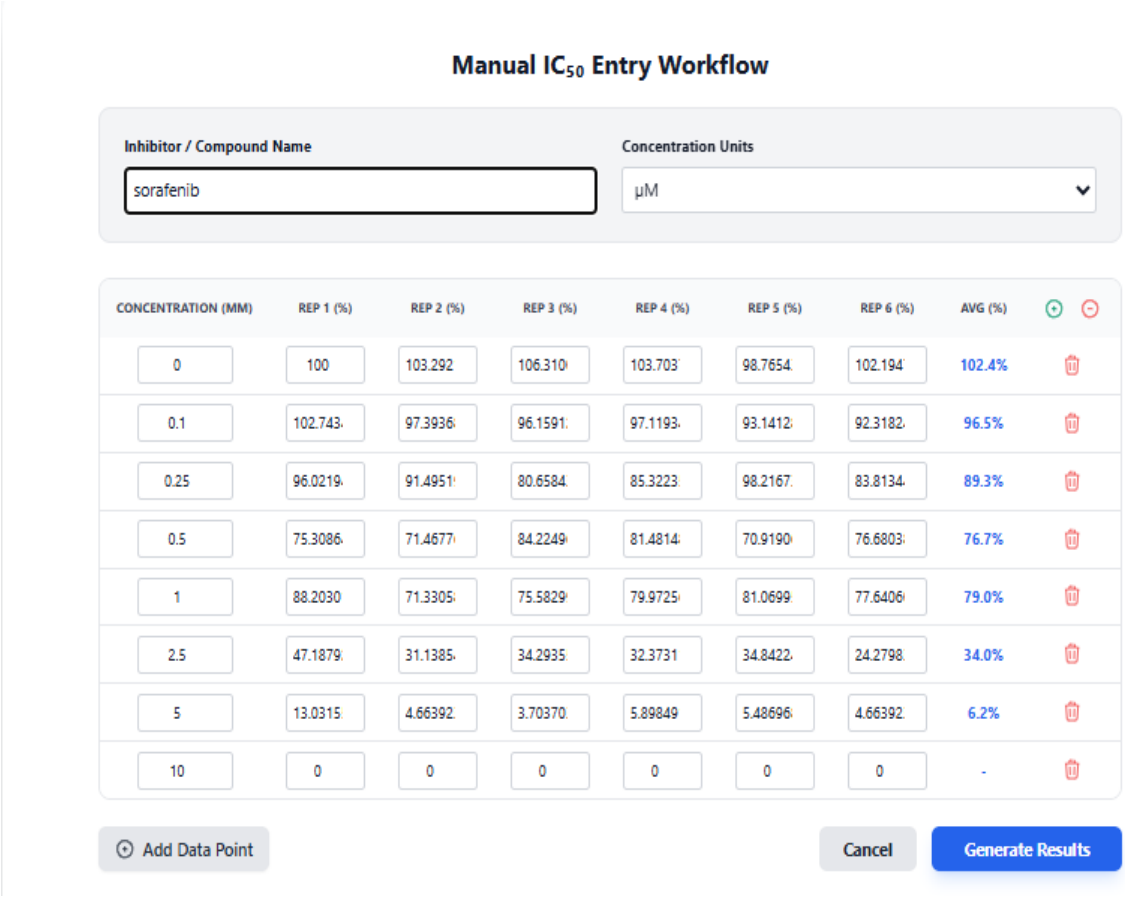
You can customise the data table using the buttons provided:

- Add Data Point button (bottom-left): Adds a new row (concentration level) to the table.
- Trash icon (right of each row): Removes that concentration row from the table. At least one row must remain.
- Green + button (in header): Adds a new replicate column to ALL rows simultaneously.
- Red − button (in header): Removes the last replicate column from all rows. Minimum one replicate must remain.



4.5 Step 5 - Generating Results

Once you have entered at least two concentration levels with viability data, click the blue "Generate Results" button at the bottom-right of the page. The application will instantly calculate statistics and IC_{50} and take you to the Results page.



5. UNDERSTANDING THE RESULTS PAGE

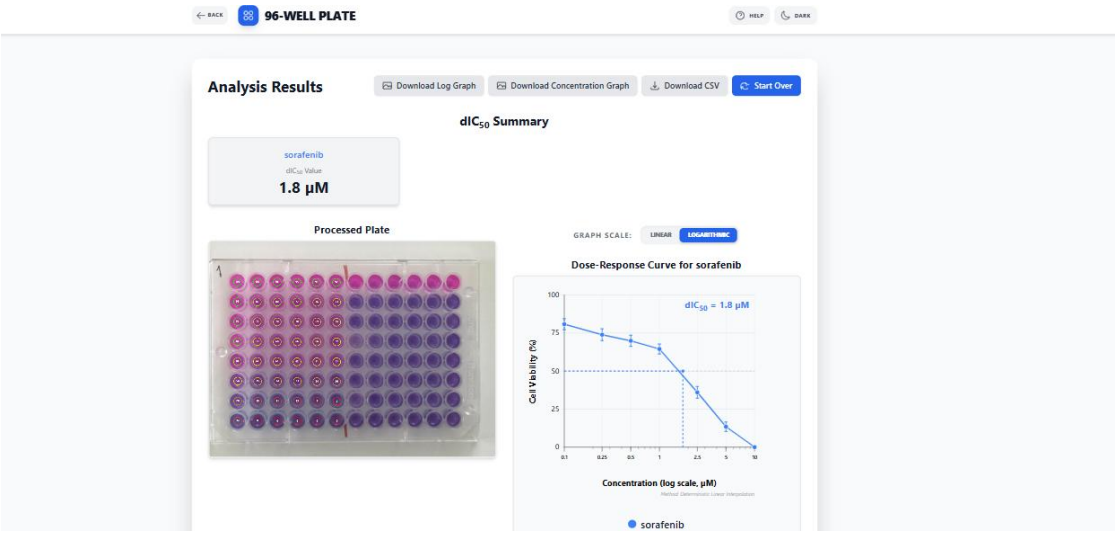
The Results page is your analytical dashboard. It brings together all computed measurements, visualisations, and export options in a single view. The page is divided into four main sections.

5.1 dIC_{50} Summary Cards

At the very top of the results page, you will see one or more summary cards, one for each inhibitor analysed. Each card displays:

- Inhibitor Name: The name you entered during setup (shown in the compound's assigned color).
- dIC_{50} Value label.
- The calculated dIC_{50} value displayed prominently in large bold text, including its unit (e.g., -

If the IC50 falls outside the concentration range you tested, the card will show "IC_{50} is outside the tested concentration range" in red text. If fewer than two data points were provided, it shows "N/A".



5.2 Processed Plate View

The left column of the results area (for dIC50 image analysis only) shows your processed plate image. On top of the photo, the application draws:

- A colored circle over each analysed well. The circle color goes from red (low viability, near 0%) through yellow to green (high viability, near 100%).
- A percentage number displayed at the center of each circle, showing the calculated cell viability for that individual well.

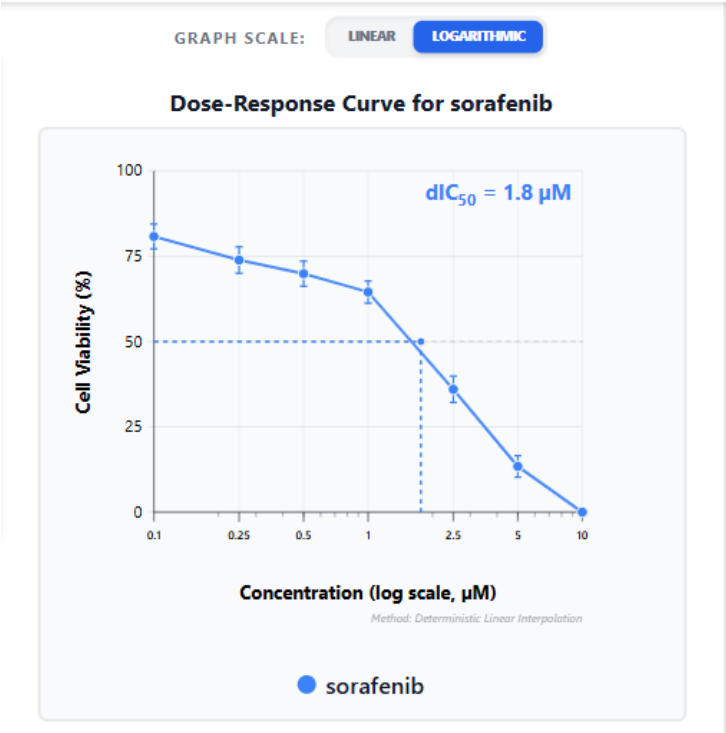
On mobile devices, a grid of colored circles replaces the image. Tap any well circle to open a magnifier view that lets you select a nearby well and see its detailed stats.

5.3 Dose-Response Curve

The right column contains the interactive dose-response graph. This is the key scientific output of the analysis.

The graph has the following features:

- X-Axis**:** Drug concentration. Can be displayed in Linear or Logarithmic scale (toggle using the buttons above the graph).
- Y-Axis**:** Cell Viability (%), ranging from 0 to 100.
- Data points: Circles (filled) for AI-analysed data, squares (outlined) for manually entered data. Error bars show the standard error of the mean.
- Connecting line: A piecewise linear interpolation curve connecting the data points.
- IC_{50} marker**:** A blue dashed crosshair marks the exact IC50 point — a vertical line from the IC_{50} concentration on the X-axis to the 50% viability level, and a horizontal line from 50% on the Y-axis to the intersection point.
- IC_{50} label**:** Displayed in the top-right corner of the graph area, showing the compound name and IC_{50} value.
- Legend: Below the graph, showing compound names with their colors.



5.4 Detailed Data Tables

Below the graph and plate view is the Detailed Dose-Response Data section. This contains a full breakdown of all measurements:

For dIC50 (image) analysis: A table with one row per concentration, columns for each replicate well's individual viability, plus the average viability and standard deviation for that concentration level.

For manual entry analysis: A table with concentration, average viability, standard deviation, and the number of replicates entered (N).

Each table row is color-alternating for easy reading. You can scroll the table horizontally if it has many replicate columns.

Detailed Dose-Response Data

sorafenib Cell Viability (%)

CONCENTRATION	REPLICATE 1	REPLICATE 2	REPLICATE 3	REPLICATE 4	REPLICATE 5	REPLICATE 6	AVG. VIABILITY (%)	STD. DEV.
0 µM	100.00	100.00	96.27	95.47	92.51	92.02	93.30	1.87
0.1 µM	97.75	81.68	80.20	77.11	75.56	72.76	80.80	8.88
0.25 µM	91.61	77.21	70.32	69.98	68.81	65.28	73.90	9.52
0.5 µM	87.46	70.32	68.21	65.81	66.20	61.69	69.90	9.05
1 µM	80.76	60.90	61.12	62.44	60.61	60.89	64.50	8.01
2.5 µM	54.03	35.32	33.92	33.34	33.92	25.75	36.00	9.45
5 µM	27.71	14.86	10.51	12.96	6.85	7.34	13.40	7.68
10 µM	13.85	0.00	0.00	0.00	0.00	0.00	0.00	0.00

5.5 Exporting Your Results

At the top-right of the results page there are four action buttons:

- Download Log Graph**:** Saves the dose-response curve in logarithmic scale as an SVG vector graphics file. SVG files are resolution-independent and ideal for publications.
- Download Concentration Graph**:** Saves the dose-response curve with a linear X-axis as an SVG file.
- Download CSV: Exports all data as a CSV spreadsheet file (compatible with Excel, Google Sheets, etc.). Includes the IC50 summary, all dose-response data points with statistics, and the full per-well viability grid.
- Start Over: Resets the entire application and returns you to the Method Selection page so you can begin a new analysis.

6. USER INTERFACE FEATURES

6.1 Dark / Light Theme Toggle

The application supports both a dark and a light visual theme. The toggle button is located in the top-right corner of the header bar at all times. It shows a Sun icon (to switch to light mode) in dark mode, and a Moon icon (to switch to dark mode) in light mode.

The chosen theme is saved in your browser's local storage, so it persists between sessions — you do not need to re-select it every time you open the app.

6.2 Navigation and Back Button

Once you move past the Welcome page, a Back button appears at the top-left of the header bar. Clicking it takes you one step back in the workflow.

Importantly, if you navigate back during grid calibration, the back button will intelligently undo calibration steps in order (e.g., from "assign wells" back to "select bottom-left", then back to "select top-right", etc.) rather than jumping all the way to a previous page.

6.3 Help Modal

Click the "Help" button (question mark icon) in the top-right header to open the Help Modal overlay. This modal contains:

- Tutorial Video**:** An embedded video player showing a walkthrough of the application.
- User Manual Download: A button to download the official PDF user guide file (dic_{50}-user-guide.pdf). This is the document you are currently reading.

Close the modal by clicking the X button in the top-right corner of the modal, or by clicking the dark backdrop outside the modal area.

7. SCIENTIFIC BACKGROUND

7.1 How Cell Viability is Measured from Images

The core principle behind the dIC_{50} method is that cell density within a well correlates with the color/intensity of the well. Cells in culture typically form a visible layer that alters the color of the medium (e.g., due to metabolic activity changing the pH indicator dye, or cell density affecting opacity).

The application converts each well's pixel data into the CIE L*a*b* color space. This color space is designed to be perceptually uniform — meaning equal numerical distances in L*a*b* correspond to equal perceived color differences to the human eye. This makes it far more suitable for quantitative color comparison than raw RGB.

The viability calculation uses a vector projection approach:

1. The average L*a*b* color of the 0% control wells (blank/dead cells) defines the "zero viability" reference point in color space.
2. The average L*a*b* color of the 100% control wells (untreated/live cells) defines the "full viability" reference point.
3. A gradient vector is computed from 0% to 100% in L*a*b* space.
4. For each concentration well, its L*a*b* color is projected onto this gradient vector. Theprojection scalar gives the viability fraction (0 to 1), clamped within bounds.
5. A gamma correction factor of 0.75 is applied to the raw viability fraction to compensate for the non-linear response of camera sensors, aligning the digital measurements with results expected from traditional spectrophotometers.
6. A robust median-based color averaging is applied within each well's circular ROI (region of interest) to reduce the influence of outlier pixels caused by bubbles, scratches, or uneven illumination.

7.2 The IC50 Calculation Method

The application uses Deterministic Linear Interpolation to calculate IC_{50}. This method is transparent, reproducible, and does not assume any particular mathematical model (such as a sigmoidal Hill equation).

The algorithm works as follows:

1. Sort all concentration-viability pairs in ascending order of concentration.
2. Scan through adjacent pairs of data points to find the pair where viability crosses the50% threshold (one point above 50%, next point below 50%, or vice versa).
3. Apply linear interpolation within that segment to find the exact concentration at whichviability equals exactly 50%.

4. If no crossing of 50% exists within the tested range, the IC50 is reported as "outside thetested range" and the user is advised to test additional concentrations.

7.3 Understanding the Dose-Response Curve

The dose-response curve plots drug concentration (X-axis) against cell viability (Y-axis). Typically, as concentration increases, viability decreases — forming a sigmoidal (S-shaped) curve when plotted on a logarithmic concentration axis.

Key features to look for in your curve:

- High plateau at low concentrations: Viability near 100% confirms your untreated control and low-dose wells are consistent.
- Low plateau at high concentrations: Viability near 0% at the highest dose confirms the drug is effective at those levels.
- Steep transition region: The most informative part of the curve. The IC50 lies within this region.
- Error bars: Small error bars indicate consistent replicates. Large error bars suggest high variability among replicate wells at that concentration.